

Name of Student:

Sch. No.

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL
Department of Electrical Engineering
Mid Term Examination March 2025

Course: B. Tech.

Semester: IV

Subject code: EE-224

Sub: Generation of Electrical Power

Time: 1 hr & 30 minutes

Max. Marks: 20

Course Coordinators: Dr. N. P. Patidar and Dr. Ch Hemanth

Note: Attempt all questions. Assume missing data if necessary.

I.	a) Explain various types of tariffs for charging electricity consumption to customers.	03	CO1
	b) Suggest and describe a tariff for residential consumers in the area where generation is in surplus.	01	
II.	A steam plant having an installed capacity of 200 MW is to be set up. The investment on the plant is Rs 9000 per kW of installed capacity. The useful life of plant may be taken as 20 years and salvage value of plant is 25% of initial cost. Find the annul depreciation reserve by (a) straight line method (b) sinking find method if interest rate is 7%.	04	CO3
III	Explain the various factors which determine the location of a steam power station.	04	CO2
IV	Explain the function of each component of demand side management.	04	CO1
V	A bank of capacitors each $40 \mu F$ is used to improve the power factor of a 3-phase, 400 V, 40 A load at 0.8 lagging power factor. Find the overall power factor and the ratio of new line loss to original line loss if the capacitors are connected in (a) star (b) Delta	04	CO3

END